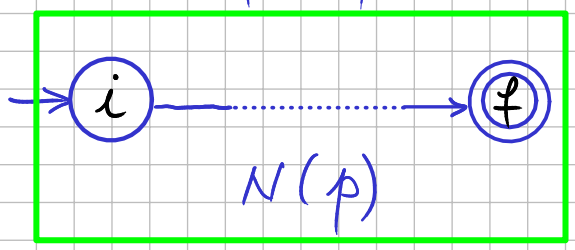
331 CC

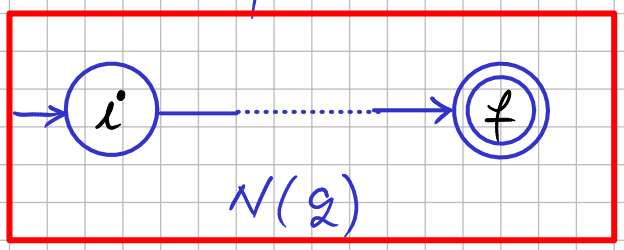
Limbaje Formale și Automate
Seminarul 6

Construcția unui AFN pornind de la o ER

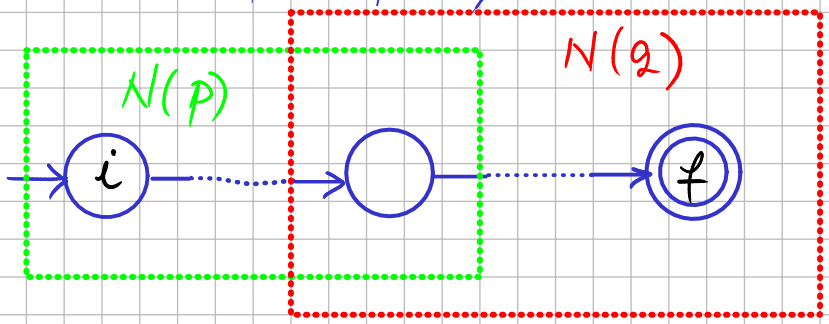
AFN pt. p



AFN pt. q

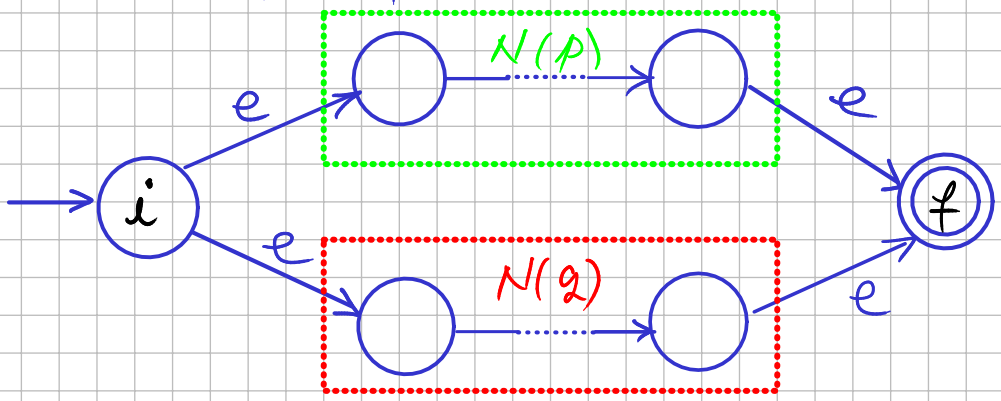


AFN pt (p q)



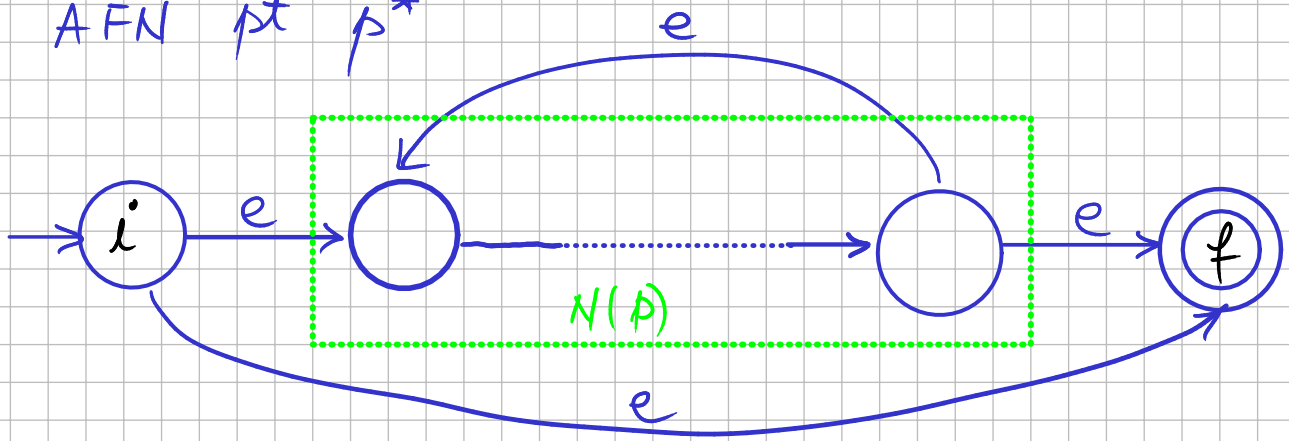
N(pq)

AFN pt (p+q)



N(p+q)

AFN pt p^*



$N(p^*)$

Algoritmul de conversie din AFN în AFD

$Q_0 = e\text{-}\hat{i}nchidere(\{q_i\})$

$$e\text{-}\hat{i}nchidere(Q') = \bigcup_{q \in Q'} e\text{-}\hat{i}nchidere(\{q\})$$

$$e\text{-}\hat{i}nchidere(\{q\}) = \begin{cases} \{q\}, \text{dacă } \nexists \text{ e-tranziții din starea } q \Rightarrow m(q,e) = \emptyset \\ \{q\} \cup \left(\bigcup_{q' \in m(q,e)} e\text{-}\hat{i}nchidere(\{q'\}) \right), \text{dacă } \exists \text{ e-tranziții din starea } q \Rightarrow m(q,e) \neq \emptyset \end{cases}$$

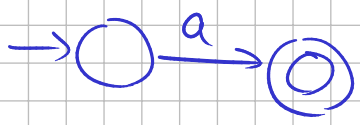
1. Construiți un AFD care acceptă limbajul descris prin ER

$(a+b)^* a^* b$

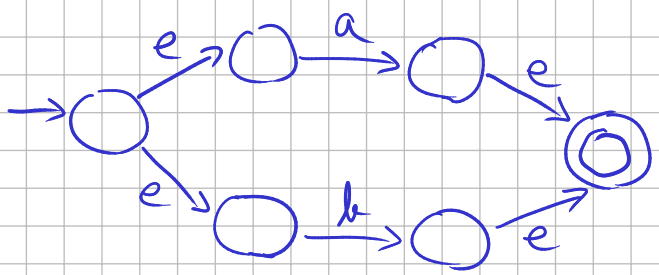
Varianta 1: ER → AFN → AFD

Trecerea în AFN

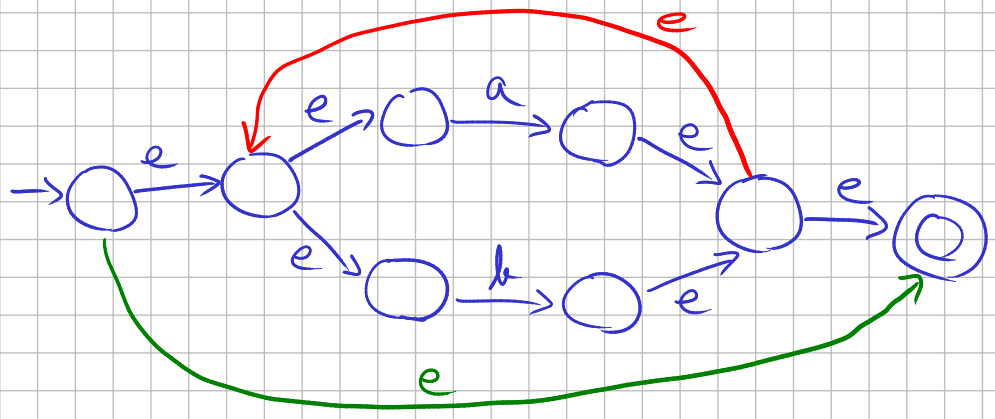
AFN pt. a



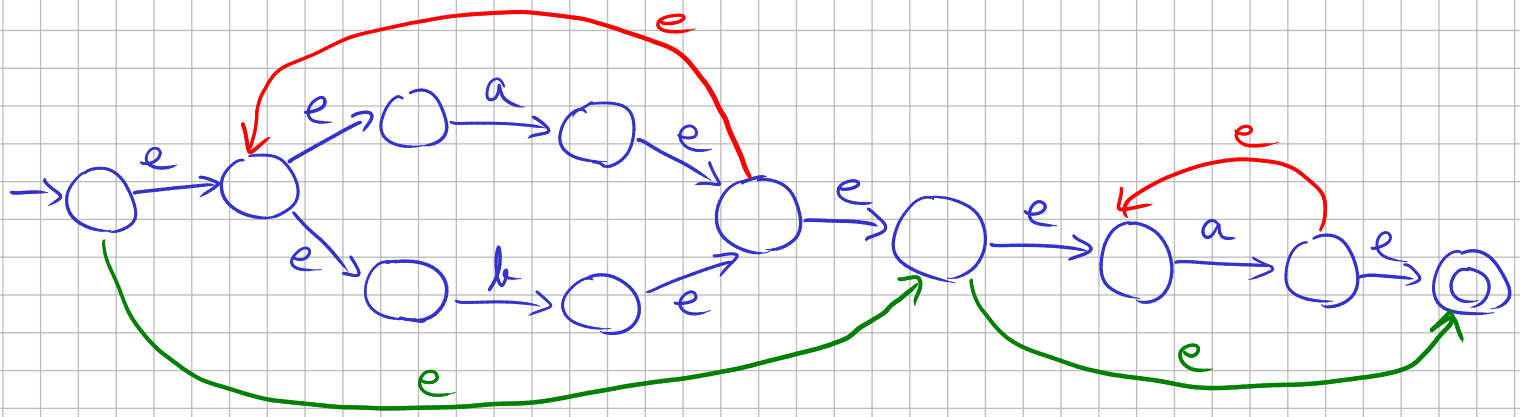
AFN pt. ER a+b



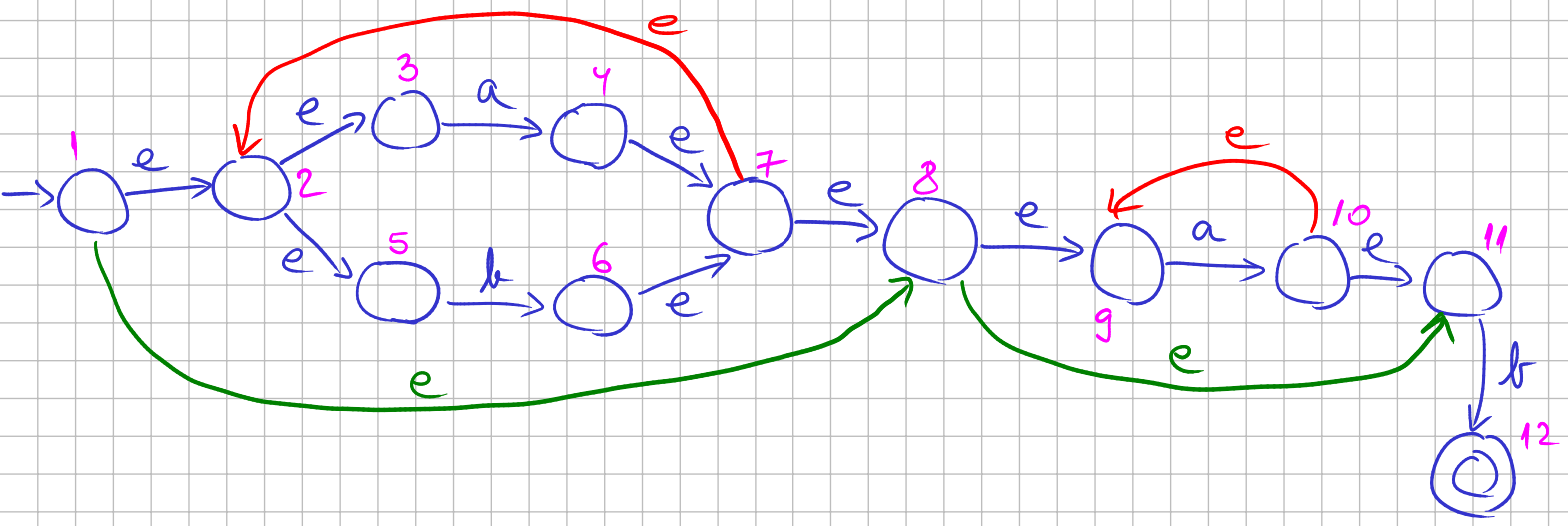
AFN pt. ER $(a+b)^*$



AFN pt. ER $(a+b)^*a^*$



AFN pt. ER $(a+b)^*a^*b$



Conversie din AFN în AFD

$Q_0 = e_închidere(\{1\}) = \{1\} \cup e_închidere(\{2,8\}) = \{1\} \cup e_închidere(\{2\})$
$\cup\ e_închidere(\{8\}) = \{1\} \cup \{2\} \cup e_închidere(\{3,5\}) \cup \{8\} \cup e_închidere(\{9,11\})$

$e_închidere(\{2\})$ $\qquad$ $e_închidere(\{8\})$

$Q_0 = \{1,2,8\} \cup e_închidere(\{3\}) \cup e_închidere(\{5\}) \cup e_închidere(\{9\}) \cup e_închidere(\{11\}) =$

$= \{1,2,8\} \cup \{3\} \cup \{5\} \cup \{9\} \cup \{11\} = \{1,2,3,5,8,9,11\}$

$m(Q_0, a) = e_închidere(mutare(Q_0, a)) = e_\hat{\imath}(\{4,10\}) =$

$= e_\hat{\imath}(\{4\}) \cup e_\hat{\imath}(\{10\}) = \{4,7,2,3,5,8,9,11\} \cup \{10,9,11\} =$

$e_\hat{\imath}(\{4\})$ $\quad$ $e_\hat{\imath}(\{10\})$

$= \{2,3,4,5,7,8,9,10,11\} = Q_1$

$m(Q_0, b) = e_\hat{\imath}(mutare(Q_0, b)) = e_\hat{\imath}(\{6,12\}) = e_\hat{\imath}(\{6\}) \cup$

$\cup e_\hat{\imath}(\{12\}) = \{6,7,2,3,5,8,9,11\} \cup \{12\} =$

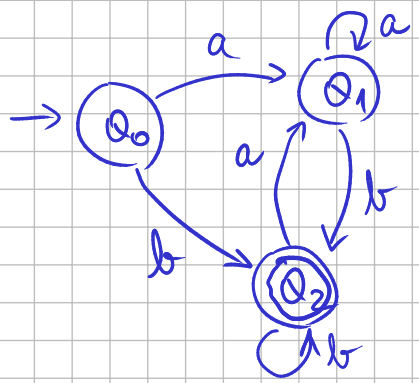
$= \{2,3,5,6,7,8,9,11,12\} = Q_2$

$m(Q_1, a) = e_\hat{\imath}(mutare(Q_1, a)) = e_\hat{\imath}(\{4,10\}) = Q_1$

$m(Q_1, b) = e_\hat{\imath}(mutare(Q_1, b)) = e_\hat{\imath}(\{6,12\}) = Q_2$

$m(Q_2, a) = e_\hat{\imath}(mutare(Q_2, a)) = e_\hat{\imath}(\{4,10\}) = Q_1$

$m(Q_2, b) = e_\hat{\imath}(mutare(Q_2, b)) = e_\hat{\imath}(\{6,12\}) = Q_2$



AFD

$Q_0, Q_1 \notin F$

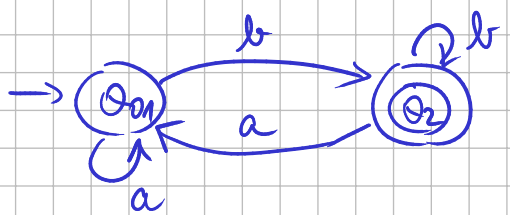
$m(Q_0, a) = Q_1$ $\quad$ $m(Q_1, a) = Q_1$

$m(Q_0, b) = Q_2$ $\quad$ $m(Q_1, b) = Q_2$

$\Downarrow$

$Q_0 \equiv Q_1$

AFD se poate rescrie



$(a+b)^* a^* b \equiv (a+b)^* b$

Varianta 2 : ER → AFD

$(a+b)^* a^* b \#$
1 2 3 4 5

simb	index	followpos
a	1	1,2,3,4
b	2	1,2,3,4
a	3	3,4
b	4	5
#	5	—

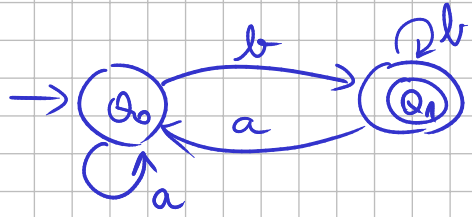
$Q_0 = firstpos(ER) = \{1,2,3,4\}$

$m(Q_0, a) = followpos(\{1,3\}) = fp(\{1\}) \cup fp(\{3\}) =$
$= \{1,2,3,4\} \cup \{3,4\} = \{1,2,3,4\} = Q_0$

$m(Q_0, b) = followpos(\{2,4\}) = fp(\{2\}) \cup fp(\{4\}) = \{1,2,3,4\} \cup \{5\} =$
$= \{1,2,3,4,5\} = Q_1$

$m(Q_1, a) = fp(\{1,3\}) = Q_0$

$m(Q_1, b) = fp(\{2,4\}) = Q_1$



2. Propuneți un AFD care acceptă limbajul descris prin ER : $(aa+bb)^* a^+ b^* \#$
1 2 3 4 5 6 7

simb	index	fp
a	1	2
a	2	3,5,1
b	3	4
b	4	3,1,5
a	5	5,6,7
b	6	6,7
#	7	—

$(aa+bb)^* a^+$
1 2 1 2 5
a a a a a
$(aa+bb)^*$

$(aa+bb)^* a^+ b^*$
5 6 6
a b b

$Q_0 = firstpos(ER) = \{1,3,5\}$

$m(Q_0, a) = fp(\{1,5\}) = fp(\{1\}) \cup fp(\{5\}) = \{2\} \cup \{5,6,7\} = \{2,5,6,7\} = Q_1$

$m(Q_0, b) = fp(\{3\}) = \{4\} = Q_2$

$m(Q_1, a) = fp(\{2,5\}) = \{1,3,5\} \cup \{5,6,7\} = \{1,3,5,6,7\} = Q_3$

$m(Q_1, b) = fp(\{6\}) = \{6,7\} = Q_4$

$m(Q_2, a) = \emptyset$

$m(Q_2, b) = fp(\{4\}) = \{1,3,5\} = Q_0$

$m(Q_3, a) = fp(\{1,5\}) = Q_1$

$m(Q_3, b) = fp(\{3,6\}) = \{4\} \cup \{6,7\} = \{4,6,7\} = Q_5$

$m(Q_4, a) = \emptyset$

$m(Q_4, b) = fp(\{6\}) = Q_4.$

$m(Q_5, a) = \emptyset$

$m(Q_5, b) = fp(\{4,6\}) = \{1,3,5\} \cup \{6,7\} = \{1,3,5,6,7\} = Q_3$

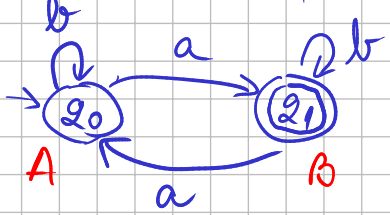
3. Propuneți un AFD care acceptă limbajul:

$L = \{ w \in \{a,b\}^* \mid \#_a(w) \text{ este impar și } \#_b(w) \text{ este par} \}$

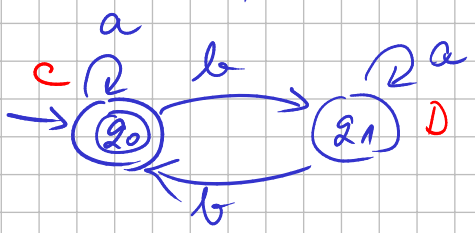
$= L_1 \cap L_2$ unde $L_1 = \{ w \in \{a,b\}^* \mid \#_a(w) \text{ este impar} \}$

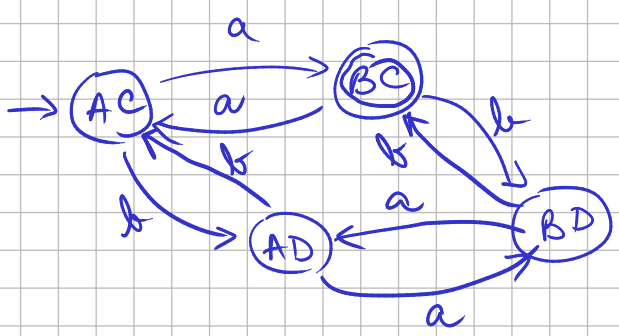
$L_2 = \{ w \in \{a,b\}^* \mid \#_b(w) \text{ este par} \}$

AFD pt. L_1

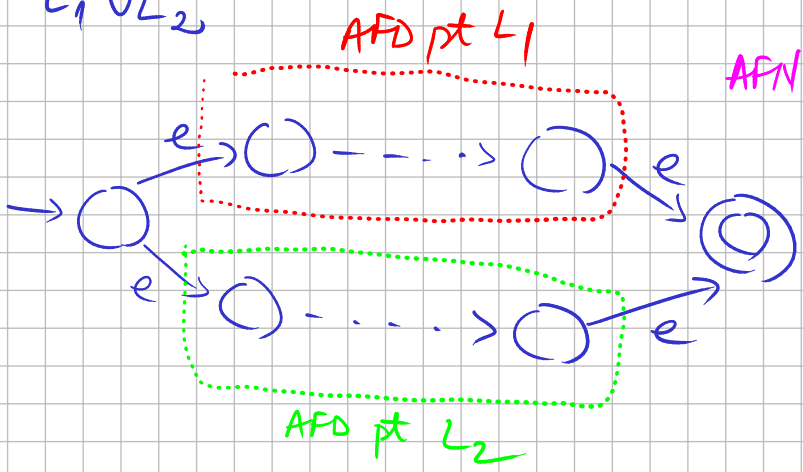


AFD pt. L_2





$L_1 \cup L_2$



AFN $\xrightarrow{\text{e-închidere}}$ AFD

Varianta 1

$\overline{L_1}$ – complementul lui L_1

$$L_1 \cup L_2 = \overline{\overline{L_1} \cap \overline{L_2}}$$

Varianta 2

$$\neg(p \vee q) = \neg p \wedge \neg q$$

$$p \vee q = \neg(\neg(p \vee q)) = \neg(\neg p \wedge \neg q)$$